

ELL8299/COL8383/COL873: Trustworthy Large Language Models

Quiz 1 – Answer Key (Semester I, 2026-27)

Total Marks: 20

Time: 60 Minutes

17/08/2026

1. **Language Modelling.** Modern language models rely on a foundational pipeline to process and generate text. Understanding their behavior requires examining how input text is segmented via tokenization, how architectures like multi-head attention capture contextual dependencies, and how metrics such as perplexity quantify the model's predictive uncertainty.

(a) Consider a vocabulary V containing K words. If a unigram language model assigns a uniform probability distribution $P(w) = 1/K$ to every word in V , what is the perplexity of any test sequence W of length M ? [2]

Solution: For $W = (w_1, \dots, w_M)$,

$$P(W) = \prod_{i=1}^M P(w_i) = \left(\frac{1}{K}\right)^M$$

The perplexity is

$$\text{PPL}(W) = P(W)^{-1/M} = \left[\left(\frac{1}{K}\right)^M\right]^{-1/M} = K$$

(b) Why is subword tokenization (e.g., BPE, WordPiece) preferred over pure word-level tokenization for LLMs? [2]

Solution: Subword tokenization can represent rare or unseen words by decomposing them into known subword units, thereby greatly reducing the OOV problem, while keeping the vocabulary size and sequence length manageable.

(c) Consider a multi-head attention layer with fixed embedding dimension d , equally divided across h heads. Would increasing h increase the total number of trainable parameters in the attention block? Explain why or why not. [1]

Solution: No. Each head has dimension $d_h = d/h$. Across all heads, the query, key, and value projections are still equivalent to matrices $W_Q, W_K, W_V \in \mathbb{R}^{d \times d}$, and the output projection satisfies $W_O \in \mathbb{R}^{d \times d}$. Thus, ignoring biases, the attention block contains $3d^2 + d^2 = 4d^2$ trainable weights, which is independent of h . Increasing h changes how the fixed dimension d is partitioned, not the total parameter count.

2. **Parameter-Efficient Fine-Tuning.** As pretrained language models scale to billions of parameters, full fine-tuning becomes increasingly expensive in terms of computation and memory. Parameter-Efficient Fine-Tuning (PEFT) reduces this cost by training only a small number of additional parameters. One widely used PEFT method is Low-Rank Adaptation (LoRA). Consider a weight matrix $W \in \mathbb{R}^{8192 \times 8192}$. In LoRA, the original matrix W is kept frozen, and its task-specific update is parameterized as $W' = W + BA$, where $A \in \mathbb{R}^{r \times 8192}$ and $B \in \mathbb{R}^{8192 \times r}$. Assume $r = 16$.

(a) How many parameters would a full fine-tuning update require for this matrix? [1]

Solution: $8192 \times 8192 = 67,108,864$ trainable parameters.

(b) How many trainable parameters does LoRA introduce? [1]

Solution: $|A| = 16 \times 8192 = 131,072$, $|B| = 8192 \times 16 = 131,072$. $|A| + |B| = 262,144$ trainable parameters.

(c) Approximately what fraction of the original parameters are trainable under LoRA? [1]

Solution:

$$\frac{262,144}{67,108,864} = \frac{1}{256} \approx 0.00390625 \times 100 \approx 0.390625\%.$$

(d) Since BA is the update applied to W , what is the maximum possible rank of the update matrix BA ? How does this compare with the maximum possible rank of W ? [2]

Solution: Using $\text{rank}(BA) \leq \min\{\text{rank}(B), \text{rank}(A)\}$, and since the inner dimension is $r = 16$,

$$\boxed{\text{rank}(BA) \leq 16}.$$

Thus the maximum possible rank of the LoRA update is 16. In contrast, for $W \in \mathbb{R}^{8192 \times 8192}$, we have

$$\boxed{\text{rank}(W) \leq 8192}.$$

Hence, the LoRA update can have maximum rank 16, whereas the original matrix can have maximum rank 8192.

3. **Memorization in LLMs.** The secret sharer setup, as discussed in class, evaluates unintended memorization by inserting random canaries $s[r]$ drawn from a randomness space $\mathcal{R}$. The exposure metric is defined as:

$$\text{exposure}_\theta(s[r]) = \log_2 |\mathcal{R}| - \log_2 \text{rank}_\theta(s[r]).$$

Assume an adversary plants a canary with a fixed prefix followed by a random 6-character alphanumeric secret (using 26 lowercase letters and 10 digits). The target model with parameters θ is evaluated, and exactly 4,095 other candidate secrets from the randomness space $\mathcal{R}$ have a strictly lower perplexity than the true planted secret $s[r]$.

(a) Calculate the value of $\text{exposure}_\theta(s[r])$. Show your intermediate calculations. *You may leave the final answer as an exact logarithmic expression if it cannot be simplified further.* [3]

Solution: There are $26 + 10 = 36$ possible characters at each of the six positions. Therefore, $|\mathcal{R}| = 36^6$. Since exactly 4095 candidates have strictly lower perplexity than the true canary, its rank is $\text{rank}_\theta(s[r]) = 4095 + 1 = 4096 = 2^{12}$. Hence,

$$\begin{aligned}\text{exposure}_\theta(s[r]) &= \log_2(36^6) - \log_2(4096) \\ &= 6 \log_2(36) - 12.\end{aligned}$$

Since $36 = 2^2 3^2$,

$$6 \log_2(36) - 12 = 12 \log_2 3.$$

Thus,

$$\text{exposure}_\theta(s[r]) = 12 \log_2 3 \approx 19.02 \text{ bits}.$$

(b) Two canaries s_1 and s_2 are drawn from the same randomness space $\mathcal{R}$. Suppose the model ranks them as $\text{rank}_\theta(s_1) = 2^{10}$ and $\text{rank}_\theta(s_2) = 2^{14}$. Which of the following statements are **TRUE**? *Select all that apply. No partial credits.* [2]

- A. ☒ s_1 has 4 bits higher exposure than s_2 .
- B. s_2 has higher exposure because its rank is larger.
- C. ☒ The model assigns a better rank to s_1 than to s_2 .
- D. The exposure difference cannot be determined without knowing $|\mathcal{R}|$.

Solution: Because both canaries use the same randomness space,

$$\begin{aligned}\text{exposure}(s_1) - \text{exposure}(s_2) &= -\log_2(2^{10}) + \log_2(2^{14}) \\ &= 14 - 10 \\ &= 4.\end{aligned}$$

Thus, s_1 has 4 bits higher exposure. Also, $2^{10} < 2^{14}$, so s_1 has the better (smaller) rank.

4. **Privacy through Re-identification.** Removing direct identifiers does not necessarily prevent re-identification, since combinations of other attributes may uniquely identify individuals. A dataset satisfies k -anonymity with respect to a set of quasi-identifiers if every combination of quasi-identifier values appearing in the dataset is shared by at least k records. The records are partitioned into equivalence classes according to their combinations of quasi-identifier values. A university releases the following supposedly anonymized dataset. Assume that {Age, Gender, Department, and City} are quasi-identifiers, while {Academic Support/Award} is a sensitive attribute.

Row	Age	Gender	Department	City	Academic Support/Award
1	28	F	CSE	Delhi	Industry Fellowship
2	34	F	CSE	Delhi	Best Thesis Award
3	31	F	CSE	Noida	Institute TAship
4	61	M	EE	Delhi	Research Assistantship
5	64	M	EE	Noida	Institute TAship
6	33	M	CSE	Delhi	Industry Fellowship

(a) With respect to the quasi-identifiers, what is the value of k for the released dataset? [1]

- A. ☒ $k = 1$
- B. $k = 2$
- C. $k = 3$
- D. $k = 6$

Solution: All six records have distinct combinations of the four quasi-identifiers. Hence each equivalence class contains one record, and $k = 1$.

(b) Suppose Age is generalized into the intervals 20–39 and 60–79, and both Delhi and Noida are generalized to “NCR”. Gender and Department remain unchanged. Which row still belongs to an equivalence class of size 1? [1]

- A. Row 1
- B. Row 3
- C. Row 5
- D. ☒ Row 6

Solution: After generalization, the equivalence classes are:

$$\begin{aligned}(20-39, F, CSE, NCR) &: \{1, 2, 3\}, \\ (60-79, M, EE, NCR) &: \{4, 5\}, \\ (20-39, M, CSE, NCR) &: \{6\}.\end{aligned}$$

Therefore, Row 6 is the only singleton equivalence class.

(c) In addition to the transformations in (b), suppose Gender is suppressed. What is the resulting value of k with respect to the remaining quasi-identifiers? [1]

A. $k = 1$

B. ✓ $k = 2$

C. $k = 3$

D. $k = 4$

Solution: After suppressing Gender, the remaining equivalence classes are

$$\begin{aligned}(20-39, CSE, NCR) &: \{1, 2, 3, 6\}, \\ (60-79, EE, NCR) &: \{4, 5\}.\end{aligned}$$

Their sizes are 4 and 2, respectively. Therefore, the minimum class size is $k = 2$.

(d) Based on the transformed dataset from (c), which of the following statements are **TRUE**? *Select all that apply.* [2]

- A. ✓ **The transformed dataset satisfies 2-anonymity with respect to the remaining quasi-identifiers.**
- B. * **Satisfying 2-anonymity guarantees that an attacker cannot infer the sensitive attribute of an individual with probability greater than $1/2$.**
- C. ✓ **Generalizing Delhi and Noida to “NCR” reduces the ability to distinguish individual records while retaining some geographic information.**
- D. Removing the sensitive attribute from the original dataset would, by itself, increase its k -anonymity from $k = 1$ to at least $k = 2$.

Solution: Statements A and C are true. The transformed dataset has minimum equivalence-class size 2, so it satisfies 2-anonymity. Generalizing the city to “NCR” reduces granularity while retaining coarse geographic information.

Statement B is false in general for 2-anonymized datasets: k -anonymity limits distinguishability using quasi-identifiers but does not guarantee diversity of sensitive values or prevent attribute disclosure.

**This option is somewhat ambiguous because the question says “based on the transformed dataset from (c).” For this particular dataset, no sensitive value occurs with probability greater than $1/2$ within either equivalence class. Therefore, credit will also be awarded to students who selected Statement B.*

Statement D is false: removing a sensitive attribute does not alter the equivalence classes formed from the quasi-identifiers, and therefore does not by itself change k .